

RDP Brute Force Attack Detection with Wazuh

In this lab, I simulate an **RDP brute force attack** in a controlled home lab environment to understand how a Security Operations Center (SOC) detects and investigates authentication attacks using Wazuh.

Lab Overview

- **Attacker Machine:** Kali Linux
- **Victim Machine:** Windows
- **Monitoring Platform:** Wazuh Manager
- **Agents:** Both Kali Linux and Windows are connected to the Wazuh Manager.

Objective

- Simulate an RDP brute force attack from Kali Linux to Windows.
- Monitor authentication events generated on the Windows system.
- Detect the attack using Wazuh alerts.
- Analyze the generated logs and identify indicators of a brute force attack.

What I Learned

- How RDP brute force attacks generate Windows security events.
- How Wazuh collects and correlates authentication logs.
- How to identify failed and successful login attempts.
- How to investigate brute force activity using SIEM alerts.
- How log analysis helps detect malicious behavior.

Real-World Relevance

- Brute force attacks are one of the most common techniques used to gain unauthorized access.
- SOC analysts continuously monitor authentication logs for repeated failed login attempts.
- SIEM platforms like Wazuh help detect, alert, and investigate these attacks in real time.
- This lab demonstrates a practical Blue Team workflow used in enterprise SOC environments.

Now, let's move on to the practical implementation of Brute force attack using the Hydra tool and detect it in the Wazuh Dashboard and Windows Event Logs

Execute below commands in the OS where your Wazuh server is running (Optional Step)

1. `systemctl restart wazuh-manager`
2. `systemctl restart wazuh-dashboard`
3. `systemctl restart wazuh-indexer`

Special privileges assigned to new login → Indicates that a user with administrative privileges has successfully logged in.

Note: Perform these security testing activities only in authorized lab environments. Never attempt them on production or third-party systems without explicit permission

1. Brute Force Attack Detection (SSH / RDP)

- **Do:**
 - Use Hydra or manual wrong password attempts
 - Target SSH (Linux) or RDP (Windows)

```
(demo@Demo)~$ hydra -l admin -P ~/Desktop/pass_list.txt rdp://192.168.198.120 -t 1 -W 5 -V
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

4. How to Improve Chances
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-03-26 10:20:15
[WARNING] the rdp module is experimental. Please test, report - and if possible, fix.
[DATA] max 1 task per 1 server, overall 1 task, 8 login tries (l:1/p:8), ~8 tries per task
[DATA] attacking rdp://192.168.198.120:3389/
[ATTEMPT] target 192.168.198.120 - login "admin" - pass "Hello" - 1 of 8 [child 0] (0/0)
[ATTEMPT] target 192.168.198.120 - login "admin" - pass "How" - 2 of 8 [child 0] (0/0)
[ATTEMPT] target 192.168.198.120 - login "admin" - pass "are" - 3 of 8 [child 0] (0/0)
[ATTEMPT] target 192.168.198.120 - login "admin" - pass "you" - 4 of 8 [child 0] (0/0)
[ATTEMPT] target 192.168.198.120 - login "admin" - pass "123" - 5 of 8 [child 0] (0/0)
[3389][rdp] host: 192.168.198.120 login: admin password: 123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-03-26 10:20:36
```

- -t 1 → 1 parallel connection
- -W 5 → wait 5 seconds between attempts
- -V → verbose (you already use this)
- **On Wazuh Dashboard:**
 - Click on the 3 horizontal lines on top left
 - Click on **Threat Hunting** under **Threat intelligence** tab
 - Multiple failed login attempts

- Rule-based alert (e.g., repeated failures)

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	timestamp	agent.name	rule.description	rule.level	rule.id
	Mar 26, 2026 @ 10:20:37.142	SamWindowsAgent	Windows User Logoff	3	60137
	Mar 26, 2026 @ 10:20:37.128	SamWindowsAgent	Successful Remote Logon Detected - User:\Admin - NTLM authentication, possible pass-the-hash ...	6	92657
	Mar 26, 2026 @ 10:20:37.127	SamWindowsAgent	Special privileges assigned to new logon.	3	67028
	Mar 26, 2026 @ 10:20:37.115	SamWindowsAgent	Special privileges assigned to new logon.	3	67028
	Mar 26, 2026 @ 10:20:36.506	SamWindowsAgent	Windows application error event.	9	60602
	Mar 26, 2026 @ 10:20:32.086	SamWindowsAgent	Logon Failure - Unknown user or bad password	5	60122
	Mar 26, 2026 @ 10:20:27.037	SamWindowsAgent	Logon Failure - Unknown user or bad password	5	60122
	Mar 26, 2026 @ 10:20:22.003	SamWindowsAgent	Logon Failure - Unknown user or bad password	5	60122
	Mar 26, 2026 @ 10:20:16.736	SamWindowsAgent	Logon Failure - Unknown user or bad password	5	60122
	Mar 26, 2026 @ 10:18:57.412	SamWindowsAgent	Service startup type was changed	3	61104
	Mar 26, 2026 @ 10:18:54.349	SamWindowsAgent	Service startup type was changed	3	61104

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W. Threat Hunting SamWindowsAgent

10:00 12:00 14:00 16:00 18:00 20:00

10:00 12:00 14:00 16:00 18:00 20:00

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	timestamp	agent.name	rule.description
	Mar 26, 2026 @ 10:20:37.142	SamWindowsAgent	Windows User
	Mar 26, 2026 @ 10:20:37.128	SamWindowsAgent	Successful Re
	Mar 26, 2026 @ 10:20:37.127	SamWindowsAgent	Special privile
	Mar 26, 2026 @ 10:20:37.115	SamWindowsAgent	Special privile
	Mar 26, 2026 @ 10:20:36.506	SamWindowsAgent	Windows appli
	Mar 26, 2026 @ 10:20:32.086	SamWindowsAgent	Logon Failure
	Mar 26, 2026 @ 10:20:27.037	SamWindowsAgent	Logon Failure
	Mar 26, 2026 @ 10:20:22.003	SamWindowsAgent	Logon Failure
	Mar 26, 2026 @ 10:20:16.736	SamWindowsAgent	Logon Failure
	Mar 26, 2026 @ 10:18:57.412	SamWindowsAgent	Service startu
	Mar 26, 2026 @ 10:18:54.349	SamWindowsAgent	Service startu
	Mar 26, 2026 @ 10:18:51.184	SamWindowsAgent	Software prot
	Mar 26, 2026 @ 10:18:30.893	SamWindowsAgent	Windows Syst
	Mar 26, 2026 @ 10:18:09.510	SamWindowsAgent	Windows Syst

Document Details

data.win.eventdata.targetuse rSid

data.win.eventdata.workstatid Demo

data.win.system.channel Security

data.win.system.computer DESKTOP-77J1CSG.shushant.com

data.win.system.eventID 4625

data.win.system.eventRecordID 36342

data.win.system.keywords 0x8010000000000000

data.win.system.level 0

data.win.system.message

Logon ID: 0x0

Logon Type: 3

Account For Which Logon Failed: Security ID: S-1-0-0

data.win.system.opcode 0

data.win.system.processID 668

data.win.system.providerGuid {54849625-5478-4994-a5ba-3e3b0328c30d}

data.win.system.providerName Microsoft-Windows-Security-Auditing

data.win.system.severityValue AUDIT_FAILURE

data.win.system.systemTime 2026-03-26T10:20:30.5966527Z

data.win.system.task 12544

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- Using the command below as well, you can detect brute force attacks in Windows (*ignore the time-stamp*)

```
PS C:\WINDOWS\system32> Get-EventLog -LogName Security -InstanceId 4625
```

Index	Time	EntryType	Source	InstanceID	Message
79228	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...
79227	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...
79226	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...
79225	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...
79224	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...
79223	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...
79222	May 22 04:05	FailureA...	Microsoft-Windows...	4625	An account failed to log on...

Outcome: Successfully detected a real-world brute force attack using Wazuh.